

## **Integer Programming Problem (IPP)**

An Integer Programming Problem is a special type of linear programming problem (LPP) **in which some or all of the decision variables are restricted to take integer values.**

### **Types of Integer Programming Problems**

- **Pure Integer Programming** – all decision variables are required to take integer values.
- **Mixed Integer Programming (MIP)** – only some decision variables are required to be integers, others may be continuous.
- **Binary Integer Programming** – decision variables are restricted to take values 0 or 1 only.

We will understand the integer programming problem by two techniques:

- **Branch and Bound Method,**
- **Gomory's Cutting Plane Method.**

## Steps for Branch and bound technique:

1. **Relax the Integer Constraints:** Solve the LP relaxation (ignore integer restrictions).
2. **Check integrality:**
  - If all variables are integers  $\rightarrow$  optimal integer solution found.
  - If not  $\rightarrow$  go to branching.
3. **Branch:** Choose a fractional variable (say  $x_i$ ) and create two subproblems by bounding it above and below its floor/ceiling.
  - $x_i \leq \lfloor x_i^* \rfloor$ .
  - $x_i \geq \lfloor x_i^* \rfloor + 1$ ,  $x_i^*$  be the fractional value of the chosen variable.
4. **Bound:** Solve LP relaxations of the subproblems to get bounds on the objective.
5. **Fathoming (Elimination of Nodes) :** Discard a branch if it is infeasible, gives worse bound than current best, or yields an integer solution.
6. **Iterate:** Repeat branching, bounding, and pruning until no active nodes remain.
7. **Stop:** The best feasible integer solution found is the **optimal solution**.

Ex:

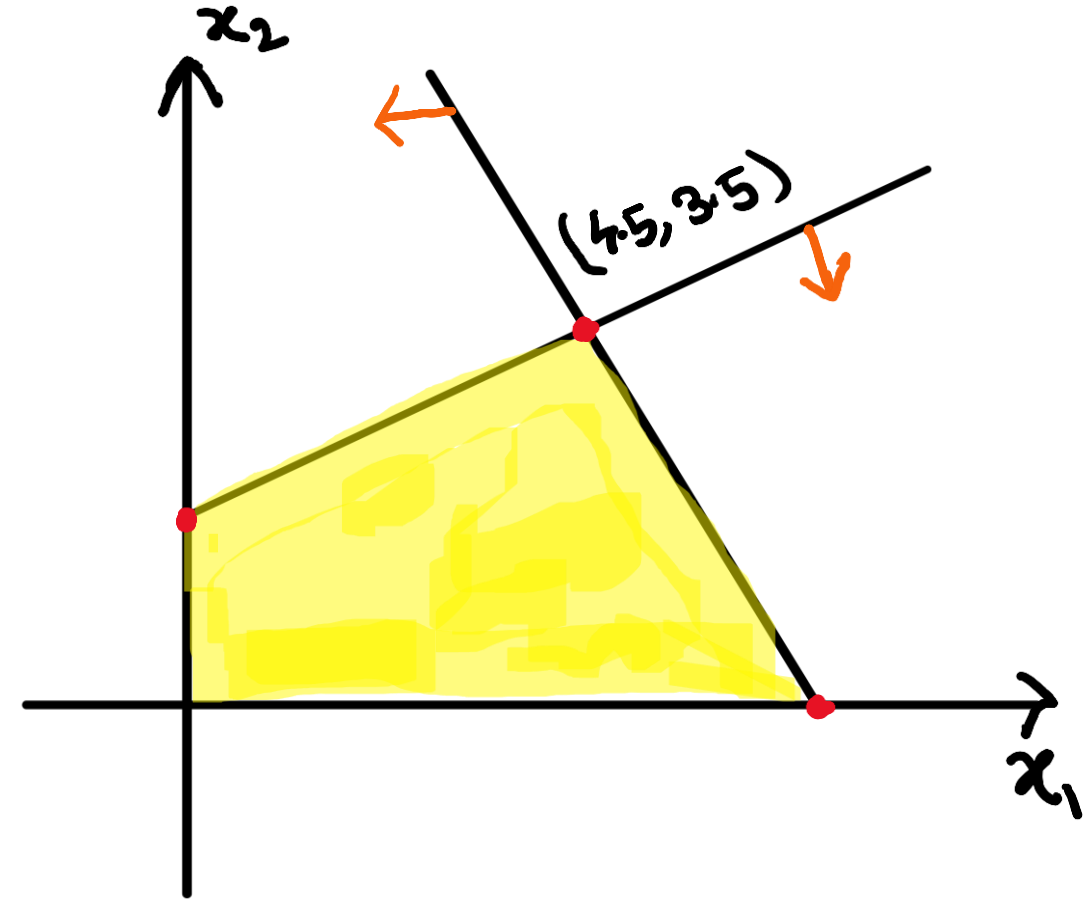
$$\begin{aligned} \text{Max } Z &= 7x_1 + 9x_2 \\ \text{Subject to: } &-x_1 + 3x_2 \leq 6, \\ &7x_1 + x_2 \leq 35, \quad x_i \geq 0, \quad x_i \in \mathbb{Z}, \end{aligned}$$

**$LP_1$  (Node 1):**  $x_1 = 4.5, x_2 = 3.5,$   
 $Z_{max} = 63$  (Upper bound)

Since, both the values are non integer, so we will start branching for the  $x_1$ , by considering two cases:

Case 1:  $x_1 \leq 4$

Case 2:  $x_1 \geq 5$



**Case 1: ( $x_1 \leq 4$ )**

**$LP_2$  (Node 2):**  $x_1 = 4$ ,  $x_2 = \frac{10}{3}$ ,  $Z_{max} = 58$  (Upper bound)

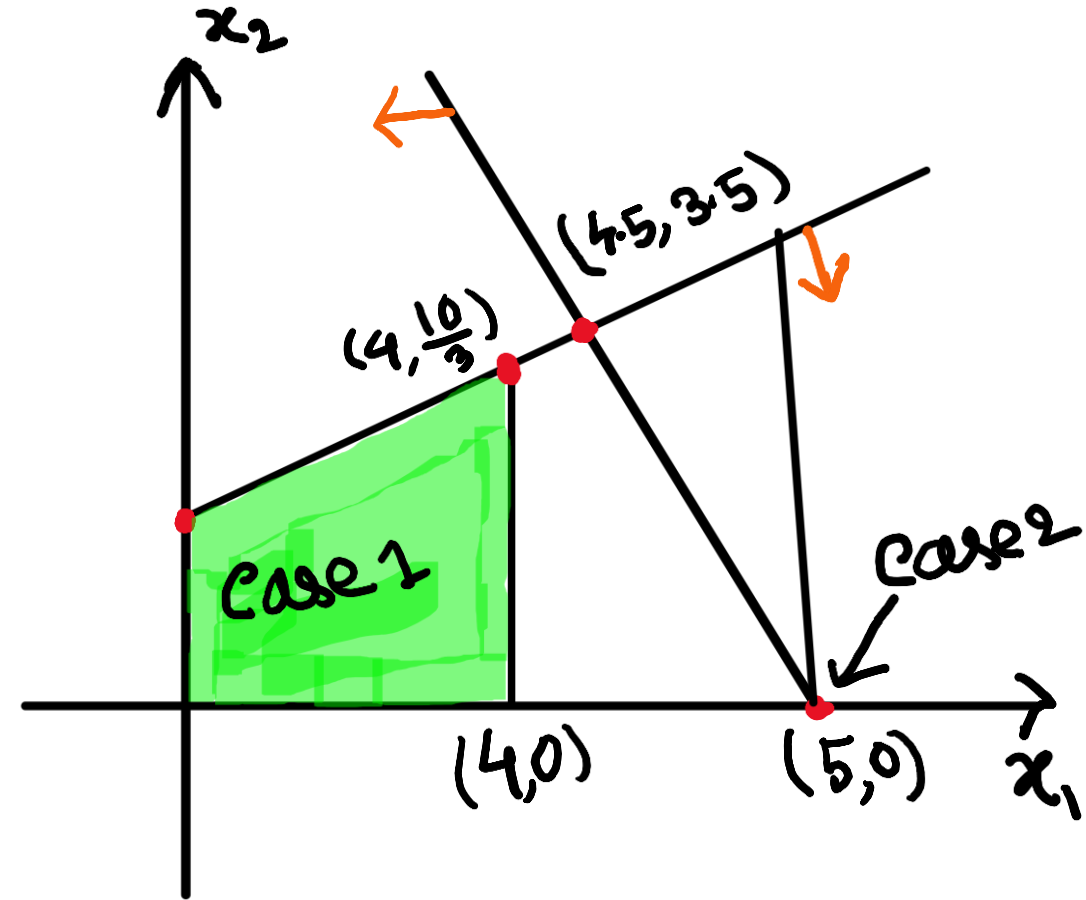
**Case 2: ( $x_1 \geq 5$ )**

**$LP_3$  (Node 3):**  $x_1 = 5$ ,  $x_2 = 0$ ,  $Z_{max} = 35$  (Lower bound)

Since, the value of  $x_1$  is non integer, so we will start branching for the  $x_2$ , by considering another two case:

Case 1.1:  $x_2 \leq 3$

Case 1.2:  $x_2 \geq 4$

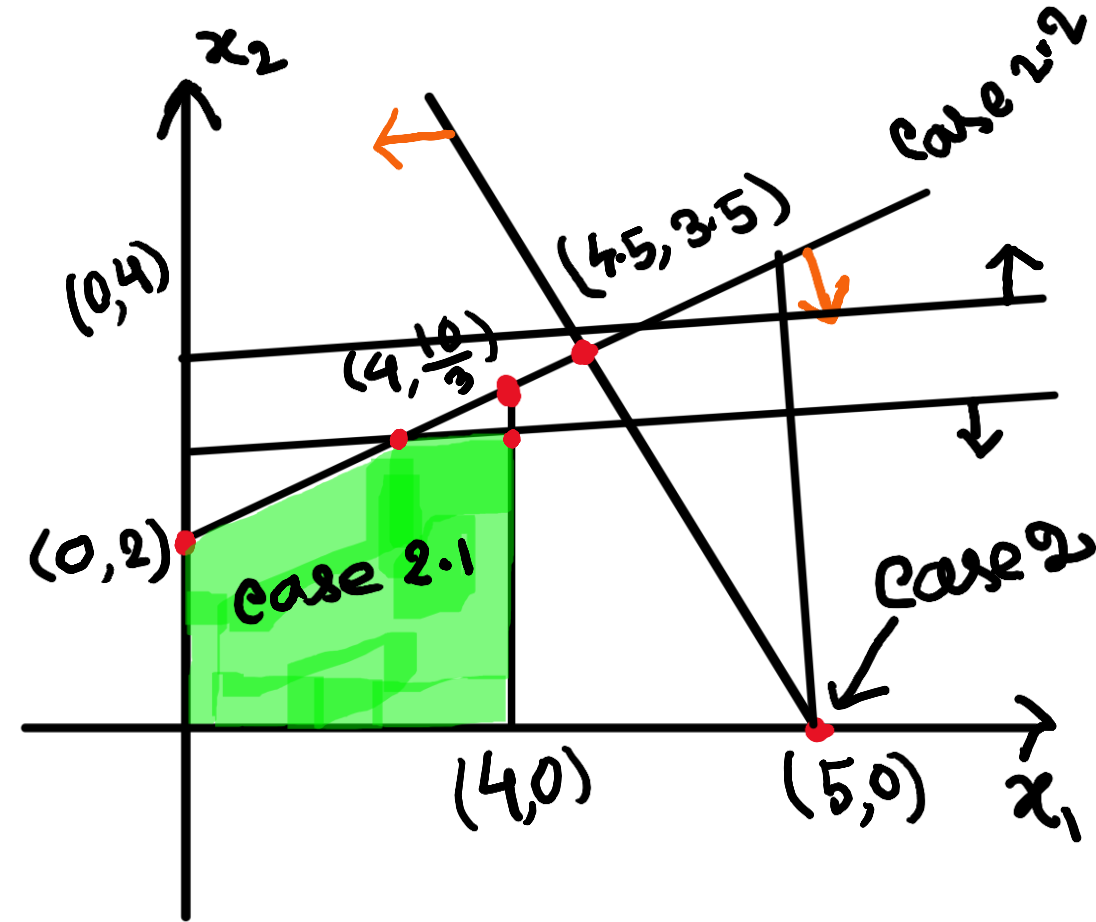


Case 1.1: ( $x_2 \leq 3$ )

**$LP_4$  (Node 4):**  $x_1 = 4$ ,  $x_2 = 3$ ,  $Z_{max} = 55$  (Current Lower bound)

Case 1.2: ( $x_2 \geq 4$ )

**$LP_5$  (Node 5):** Infeasible Solution



**$LP_1$  (Node 1):**  $x_1 = 4.5, x_2 = 3.5,$   
 $Z_{max} = 63$  (Upper bound)

**$LP_2$  (Node 2):**  $x_1 = 5, x_2 = 0, Z_{max} =$   
35 (Fathomed)

**$LP_3$  (Node 3):**  $x_1 = 4, x_2 = \frac{10}{3}, Z_{max} =$   
58 (Upper bound)

**$LP_4$  (Node 4):**  $x_1 = 4, x_2 = 3, Z_{max} =$   
55 (Current Lower bound)

**$LP_5$  (Node 5):** Infeasible Solution

Therefore, the final solution is:

$$x_1 = 4, \quad x_2 = 3, \quad Z_{max} = 55.$$

Ex:

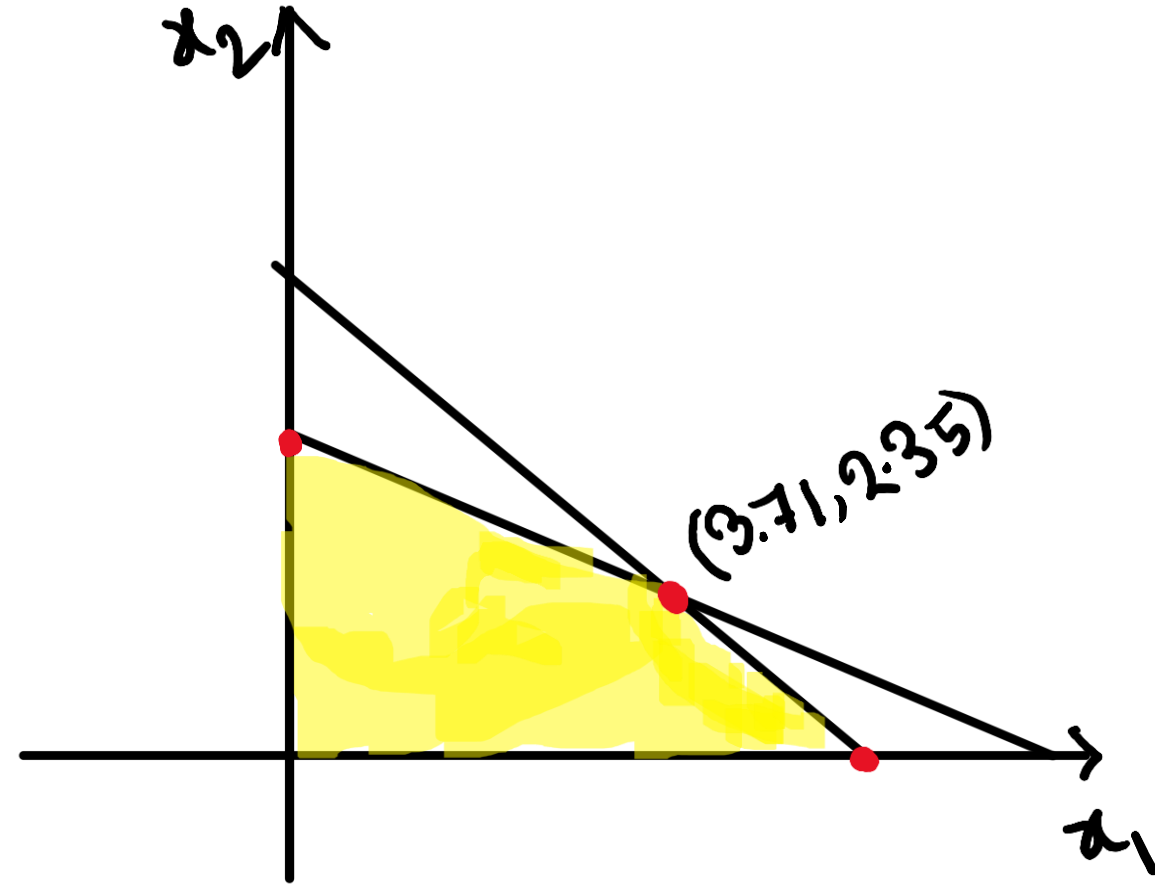
$$\begin{aligned} \text{Max } Z &= 2x_1 + 3x_2 \\ \text{Subject to: } 5x_1 + 7x_2 &\leq 35, \\ 4x_1 + 9x_2 &\leq 36, \quad x_i \geq 0, \quad x_i \in \mathbb{Z}, \end{aligned}$$

**$LP_1$  (Node 1):**  $x_1 = 3.71, x_2 = 2.35,$   
 $Z_{max} = 14.47$  (Upper bound)

Since, both the values are non integer, so we will start branching for the  $x_1$ , by considering two cases:

Case 1:  $x_1 \leq 3$

Case 2:  $x_1 \geq 4$



**Case 1: ( $x_1 \leq 3$ )**

**$LP_2$  (Node 2):**  $x_1 = 3$ ,  $x_2 = 2.66$ ,  
 $Z_{max} = 14$

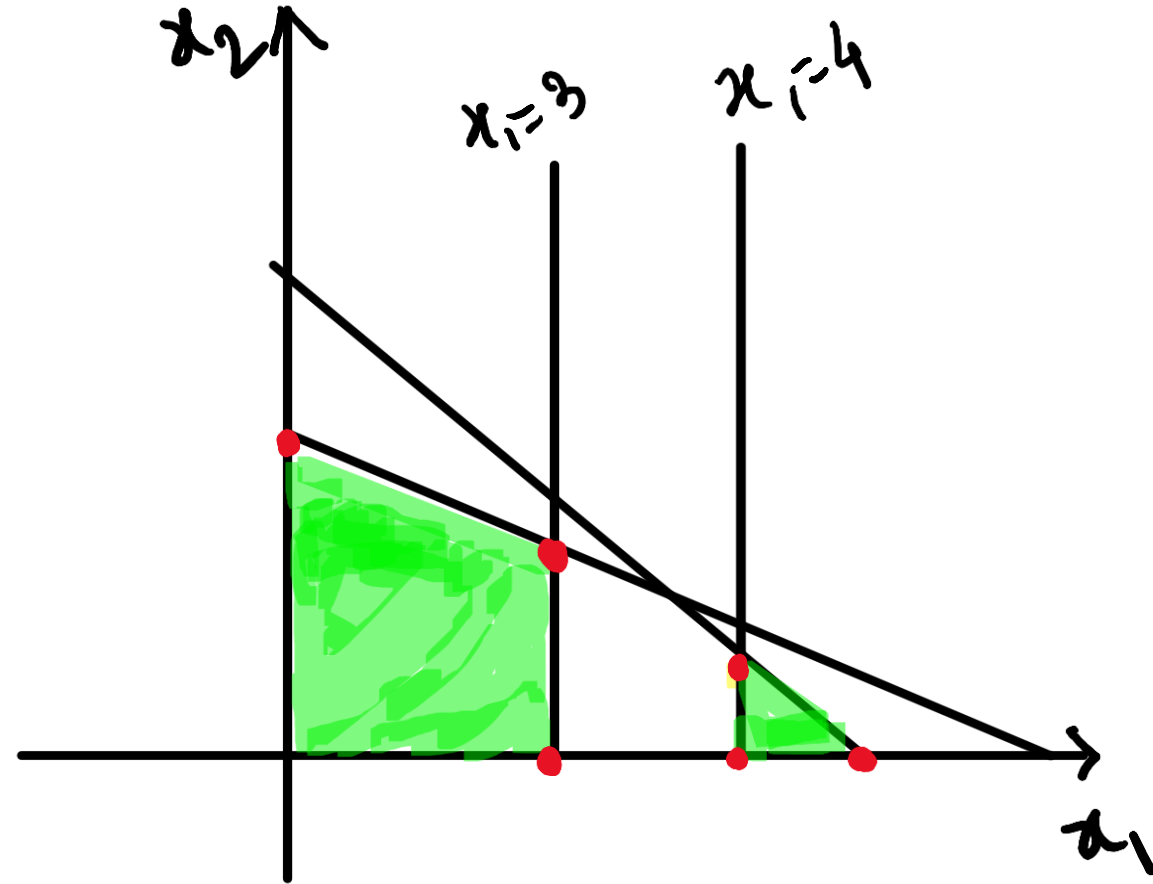
**Case 2: ( $x_1 \geq 4$ )**

**$LP_3$  (Node 3):**  $x_1 = 4$ ,  $x_2 = 2.44$ ,  $Z_{max} =$   
14.42 (Upper bound)

Since, the value of  $x_1$  is non integer, so we will start branching for the  $x_2$ , by considering another two case:

Case 2.1:  $x_2 \leq 2$

Case 2.2:  $x_2 \geq 3$





Case 2.1: ( $x_2 \leq 2$ )

**$LP_4$  (Node 4):**  $x_1 = 4.2$ ,  $x_2 = 2$   $Z_{max} = 14.4$

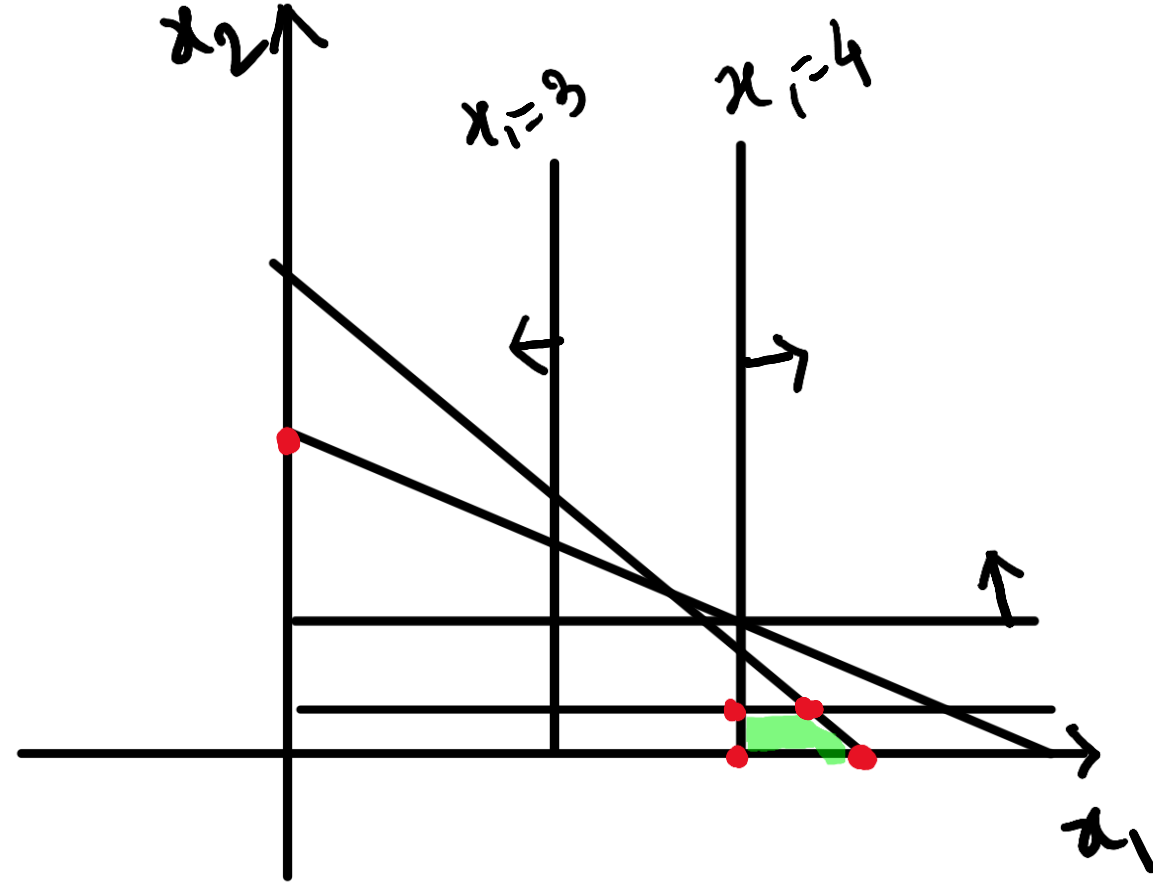
Case 2.2: ( $x_2 \geq 3$ )

**$LP_5$  (Node 5):** Infeasible Solution

Since, the value of  $x_1$  is non integer, so we will start branching for the  $x_1$ , by considering another two case:

Case 2.1.1:  $x_1 \leq 4$

Case 2.1.2:  $x_1 \geq 5$

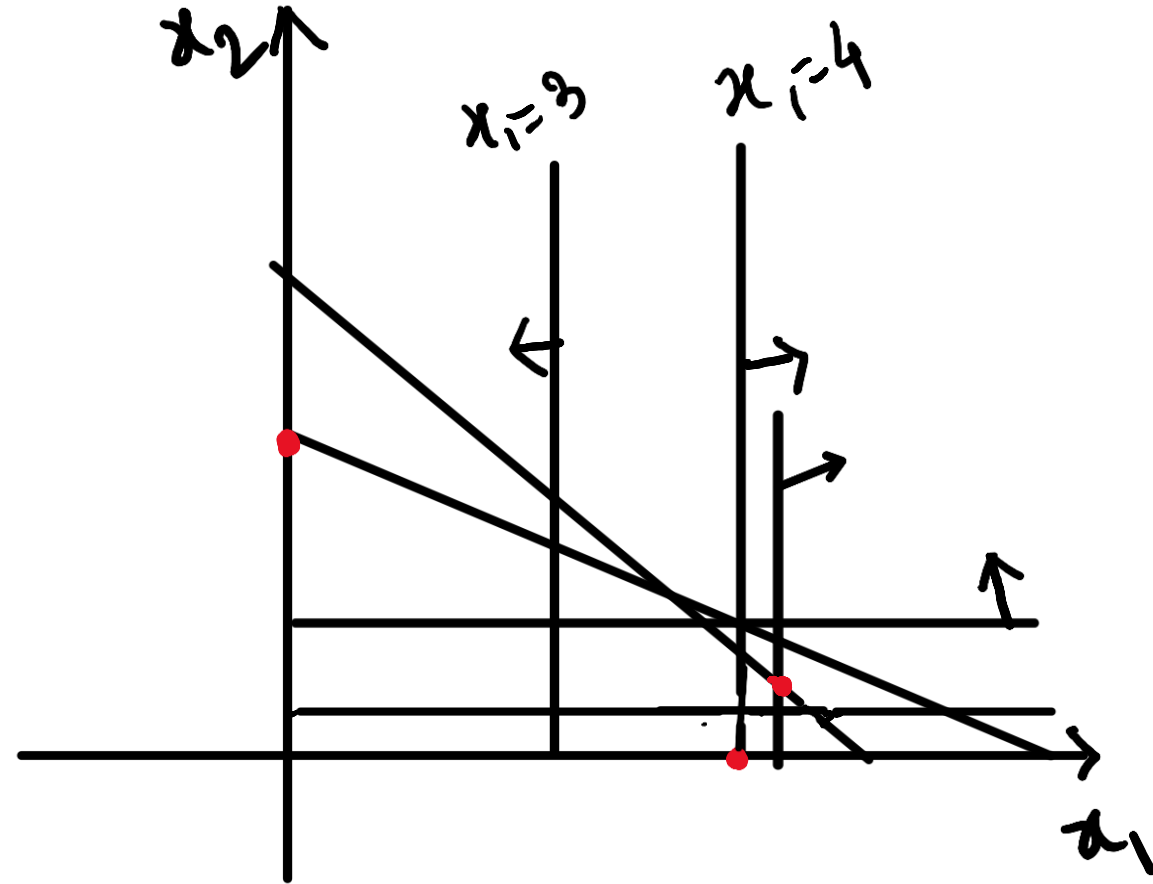


Case 2.1.1: ( $x_1 \leq 4$ )

**$LP_4$  (Node 6):**  $x_1 = 4$ ,  $x_2 = 2$ ,  $Z_{max} = 14$  (Current Lower bound)

Case 2.1.2: ( $x_1 \geq 5$ )

**$LP_5$  (Node 7):**  $x_1 = 5$ ,  $x_2 = 1.42$ ,  
 $Z_{max} = 14.28$



**$LP_1$  (Node 1):**  $x_1 = 3.71, x_2 = 2.35,$   
 $Z_{max} = 14.47$  (Upper bound)

**$LP_2$  (Node 2):**  $x_1 = 3, x_2 = 2.66,$   
 $Z_{max} = 14$  (Fathomed)

**$LP_3$  (Node 3):**  $x_1 = 4, x_2 = 2.14, Z_{max} =$   
 $14.42$  (Upper bound)

**$LP_4$  (Node 4):**  $x_1 = 4.2, x_2 = 2, Z_{max} =$   
 $14.4$

**$LP_5$  (Node 5):** Infeasible Solution

**$LP_4$  (Node 6):**  $x_1 = 4, x_2 = 2, Z_{max} =$   
 $14$  (Current Lower bound)

**$LP_5$  (Node 7):**  $x_1 = 5, x_2 = 1.42, Z_{max} =$   
 $14.28$  (Fathomed)

Therefore, the final solution is:

$$x_1 = 4, \quad x_2 = 2, \quad Z_{max} = 14.$$

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**Solve the following Integer Programming Problem by *Branch & Bound* technique:**

**Ex 1:**

$$\begin{aligned} \text{Max } Z &= 2x_1 + x_2 \\ \text{Subject to: } 10x_1 + 10x_2 &\leq 9, \\ 10x_1 + 5x_2 &\geq 1, \quad x_i \geq 0, \quad x_i \in \mathbb{Z}, \end{aligned}$$

**Ex 2:**

$$\begin{aligned} \text{Max } Z &= 3x_1 + 2x_2 \\ \text{Subject to: } 2x_1 + 5x_2 &\leq 9, \\ 4x_1 + 2x_2 &\leq 9, \quad x_i \geq 0, \quad x_i \in \mathbb{Z}, \end{aligned}$$

**Ex 3:**

$$\begin{aligned} \text{Max } Z &= 5x_1 + 7x_2 \\ \text{Subject to: } 2x_1 + x_2 &\leq 13, \\ 5x_1 + 9x_2 &\geq 41, \quad x_i \geq 0, \quad x_i \in \mathbb{Z}, \end{aligned}$$